

RoboMP-DINOv2: Prompts, Not Filters for Robust Robot Manipulation

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Abstract

Robot manipulation policies must generalize across visual distribution shifts while retaining scene context that can affect the correct action. General-purpose vision encoders are not designed specifically for visuomotor control and may be sensitive to such shifts, whereas object-centric methods often use segmentation masks as hard filters, discarding unmasked information and making segmentation errors an information bottleneck. We propose *RoboMP-DINOv2* (Robotics Mask-Prompted DINOv2), a robot-oriented full-scene vision encoder that uses masks as spatial prompts rather than visibility filters. RoboMP-DINOv2 extracts dense DINOv2 features from the complete observation, injects learned region-specific embeddings at mask-selected locations, jointly contextualizes prompted and unprompted tokens, and aggregates them into a full-scene representation for action prediction. To improve appearance generalization, we combine RoboMP-DINOv2 with masked-region color randomization (MCR), denoting the resulting training pipeline as *RoboMP-DINOv2-MCR*. MCR randomizes selected object regions during training while preserving contextual cues elsewhere in the scene. Across seven simulated manipulation settings, RoboMP-DINOv2 achieves macro-average success rates of 60.7% under spatial shifts and 59.7% under scene clutter, compared with 50.7% and 41.0%, respectively, for a DINOv2-based Diffusion Policy. Under appearance shifts involving unseen object colors, RoboMP-DINOv2-MCR achieves 72.5% average success, compared with 35.1% for the strongest evaluated color-randomized baseline. Experiments involving obstacle avoidance and decisions based on unmasked color cues further demonstrate that the proposed designs improve robustness without discarding behaviorally relevant scene information. Paired-observation representation analyses additionally show lower sensitivity to the evaluated clutter and appearance perturbations. Together, these results establish RoboMP-DINOv2 and RoboMP-DINOv2-MCR as effective full-scene representation designs for robust robot manipulation under the visual shifts considered in this work. Project code is available at https://github.com/han20192019/RoboMP_DINOv2.

1 Introduction

Robot policies deployed beyond their training environments inevitably encounter visual distribution shifts. The same manipulation problem may appear against a different background, among unfamiliar objects, or with task objects of unseen colors. A central challenge is that the magnitude of a visual change does not necessarily reflect its behavioral relevance: a distractor may substantially alter an observation without changing the desired action, whereas a small obstacle entering the motion path may require a different trajectory. Effective visual representations for manipulation must therefore balance task focus with scene awareness, reducing sensitivity to nuisance variation while preserving contextual information that may affect the correct behavior.

Modern robot policies commonly rely on visual encoders pretrained for general-purpose image understanding. Models such as DINOv2 [Oquab et al., 2023] provide strong transferable features, but their pretraining objectives do not explicitly identify which image regions are relevant to a particular manipulation

problem or which visual changes should leave the policy behavior unchanged. Object-centric methods introduce such structure by using segmentation masks to isolate relevant entities in 2D images or 3D point clouds [Emukpere et al., 2025, Zhu et al., 2023]. Although this filtering can reduce sensitivity to irrelevant appearance variation, it also turns segmentation into a hard information bottleneck: pixels or points outside the selected regions are unavailable to the policy, even when they describe an obstacle, a support surface, another robot arm, or other behaviorally relevant context. Moreover, an inaccurate mask can directly remove visual evidence from the representation. These limitations motivate a different role for masks: rather than determining what the policy is allowed to observe, masks can highlight selected regions while leaving the complete visual scene accessible.

Following the principle of using masks as *prompts*, not *filters*, we propose *RoboMP-DINOv2* (Robotics Mask-Prompted DINOv2), a robot-oriented full-scene vision encoder that adapts DINOv2 to structured spatial guidance. RoboMP-DINOv2 integrates four components: dense full-scene feature extraction, multi-region mask prompting, global contextualization, and scene-level aggregation. It first extracts dense patch features from the complete RGB observation using a pretrained DINOv2 backbone. Learned region-specific embeddings are then added at locations identified by the input masks, explicitly marking selected regions without removing or replacing their underlying visual features. A lightweight transformer jointly contextualizes prompted and unprompted tokens, allowing the representation to capture interactions between selected objects and the surrounding scene. Finally, the contextualized tokens are aggregated into a compact full-scene representation for action prediction, and the complete encoder is fine-tuned jointly with the policy. Unlike hard object-centric filtering, RoboMP-DINOv2 preserves unmasked scene content and does not directly discard visual evidence when the supplied masks are imperfect. The encoder itself consumes only an ordered collection of binary masks and does not require object-category labels or textual object descriptions as inputs.

Object appearance presents a complementary source of distribution shift. Even when relevant regions are explicitly identified, a visual encoder may associate an object with the colors observed during training. Conventional full-image color randomization can weaken such correlations, but it may also modify color cues that remain behaviorally relevant, such as an indicator light that determines the correct action. The masks used by RoboMP-DINOv2 provide a natural interface for localizing this augmentation. We therefore introduce *masked-region color randomization* (MCR), which randomizes RGB appearance only within selected object regions while leaving the remainder of the scene unchanged. MCR discourages reliance on training-specific object colors while preserving scene geometry and unmasked contextual cues. Combined with the full-scene representation produced by RoboMP-DINOv2, this localized augmentation enables the policy to generalize across task-object recoloring while remaining sensitive to color information that is behaviorally relevant elsewhere in the scene.

We evaluate RoboMP-DINOv2 with Diffusion Policy [Chi et al., 2023] across seven simulated manipulation settings spanning object placement, obstacle avoidance, peg insertion, tool use, long-horizon manipulation, and bimanual stacking. Under held-out spatial configurations, RoboMP-DINOv2 achieves a macro-average success rate of 60.7%, compared with 50.7% for a standard DINOv2-based Diffusion Policy. Under scene clutter, the corresponding success rates are 59.7% and 41.0%. RoboMP-DINOv2 achieves higher observed success than DP-DINOv2 across all seven settings in both regimes and compares favorably with the evaluated 2D and 3D object-centric methods and other mask-based vision encoders. We further evaluate RoboMP-DINOv2-MCR under appearance shifts involving unseen object colors. Among the evaluated encoder and color-augmentation combinations, RoboMP-DINOv2-MCR matches the highest observed performance on one setting and outperforms all alternatives on the remaining six, achieving a macro-average success rate of 72.5%, compared with 35.1% for the strongest evaluated alternative. Experiments involving obstacle avoidance and decisions based on unmasked color cues further demonstrate that the proposed designs preserve scene information that remains relevant to control. Paired-observation analyses also show that RoboMP-DINOv2 and RoboMP-DINOv2-MCR produce representations with lower sensitivity to the evaluated clutter and recoloring perturbations when the underlying simulator state is held

fixed. We interpret these findings as evidence for the effectiveness of the complete RoboMP-DINOv2 and RoboMP-DINOv2-MCR designs under the visual shifts considered in this work.

Our contributions are twofold:

- We propose *RoboMP-DINOv2* (Robotics Mask-Prompted DINOv2), a robot-oriented full-scene vision encoder that integrates dense DINOv2 features, multi-region mask prompting, global contextualization, and scene-level aggregation for visuomotor policy learning. RoboMP-DINOv2 highlights selected regions while preserving the complete observation, thereby avoiding the hard information bottleneck imposed by object-centric filtering.
- We introduce *masked-region color randomization* (MCR) as a complementary training strategy. The resulting *RoboMP-DINOv2-MCR* pipeline improves generalization to unseen object appearances while preserving behaviorally relevant contextual cues, and achieves the strongest observed overall performance among the evaluated encoder and augmentation combinations.

2 Method

Our goal is to incorporate region-level spatial guidance into a pretrained vision encoder without restricting the policy to a predefined set of segmented regions. The proposed *RoboMP-DINOv2* (Robotics Mask-Prompted DINOv2) encoder comprises four stages: dense full-scene DINOv2 feature extraction, feature-space mask prompting, global contextualization, and scene-level aggregation. RoboMP-DINOv2 first extracts dense visual features from the complete RGB observation and then injects learned prompts at locations specified by the input masks. A lightweight transformer jointly contextualizes all prompted and unprompted tokens before aggregating them into a compact full-scene representation for the visuomotor policy. No image regions or spatial tokens are removed at any stage of the encoder.

We additionally introduce masked-region color randomization in Sec. 2.4 as an optional training augmentation for generalization to unseen object appearances. This augmentation complements RoboMP-DINOv2 but is not part of the encoder architecture itself.

2.1 Problem Formulation

At time step t , the policy receives an RGB observation $I_t \in \mathbb{R}^{3 \times H_0 \times W_0}$, robot proprioception q_t , a language instruction ℓ , and an ordered collection of R binary masks,

$$\mathcal{M}_t = \left(M_t^{(1)}, \dots, M_t^{(R)} \right), \quad M_t^{(r)} \in \{0, 1\}^{H_0 \times W_0}. \quad (1)$$

The masks identify image regions associated with manipulated objects, goals, tools, receptacles, or other selected entities. Their ordering is preserved: each mask slot r is associated with a learned prompt embedding, allowing the encoder to learn a slot-specific interpretation from demonstrations. RoboMP-DINOv2 itself receives only the ordered binary masks; object-category labels and textual object descriptions are not provided as encoder inputs.

The visual encoder g_ψ maps the RGB observation and masks to a d -dimensional full-scene representation,

$$v_t = g_\psi(I_t, \mathcal{M}_t) \in \mathbb{R}^d. \quad (2)$$

The downstream policy π_θ is conditioned on v_t , proprioception, and language. Importantly, the masks provide spatial guidance but do not remove pixels, crop objects, or otherwise restrict which regions of the observation remain available to the policy.

Mask acquisition. During training, task masks are obtained from simulator ground-truth segmentation. At evaluation time, we do not assume access to simulator masks. For each selected region, we use a text query with Grounding DINO [Liu et al., 2024] to obtain candidate bounding boxes. After non-maximum suppression, we retain the highest-confidence detection and use its bounding box to prompt SAM [Kirillov et al., 2023]. Among the masks returned by SAM, we select the highest-confidence prediction. The text queries are used only by this external mask-generation pipeline and are never provided to RoboMP-DINOv2 or the downstream policy. Before feature-space prompting, each predicted mask is resized to the spatial resolution of the visual feature map using nearest-neighbor interpolation. Thus, training uses simulator masks, whereas evaluation relies on externally predicted masks.

2.2 RoboMP-DINOv2: Mask-Prompted Full-Scene Encoding

Dense full-scene feature extraction. We resize I_t to $H_I \times W_I$ and extract patch tokens using a Vision Transformer backbone Φ initialized from DINOv2 [Oquab et al., 2023]. Let p denote the patch size, $H_f = H_I/p$, $W_f = W_I/p$, and $N = H_f W_f$. The backbone produces N patch tokens, each with dimension d_b . We restore their two-dimensional spatial arrangement and project them to the encoder dimension d using a learned 1×1 convolution W_{proj} :

$$F_t = W_{\text{proj}} (\text{reshape}(\Phi_{\text{patch}}(I_t))) \in \mathbb{R}^{d \times H_f \times W_f}, \quad (3)$$

where Φ_{patch} denotes the patch-token output of the visual backbone.

Masks are not applied during feature extraction. Consequently, F_t retains visual information from the entire scene, including selected objects, obstacles, support surfaces, the robot embodiment, distractors, and other context outside the specified mask regions.

Feature-space mask prompting. For each mask slot r , we learn a prompt embedding $e_r \in \mathbb{R}^d$. We resize $M_t^{(r)}$ to the feature resolution using nearest-neighbor interpolation, obtaining $\bar{M}_t^{(r)} \in \{0, 1\}^{H_f \times W_f}$. Let $P \in \mathbb{R}^{d \times H_f \times W_f}$ denote a learned two-dimensional positional embedding. The prompted feature at spatial location (u, v) is

$$S_t[:, u, v] = F_t[:, u, v] + P[:, u, v] + \sum_{r=1}^R \bar{M}_t^{(r)}[u, v] e_r. \quad (4)$$

Equation 4 marks mask-selected locations by adding their corresponding slot embeddings while retaining the underlying visual features. Tokens outside all masks retain their visual and positional representations. When multiple masks overlap, their prompt embeddings are added at the overlapping locations.

This operation differs from hard object masking because no spatial token is removed or replaced. It also differs from concatenating masks with the RGB input because spatial guidance is introduced directly into the pretrained feature space. The additive prompts should not be interpreted as fixed attention weights. Instead, they provide learned region-specific conditioning signals whose influence is determined by the subsequent contextualization layers and the downstream policy objective.

Global contextualization. The appropriate action can depend on interactions between prompted regions and unprompted scene context. For example, the trajectory for placing an object may depend on whether an unmasked obstacle blocks the direct path. To model such interactions, we flatten S_t into N spatial tokens and process them using a lightweight transformer encoder $\mathcal{T}$ followed by layer normalization:

$$Z_t = \text{LN}(\mathcal{T}(\text{flatten}(S_t))) \in \mathbb{R}^{N \times d}. \quad (5)$$

Self-attention allows information to propagate between prompted and unprompted tokens, enabling the encoder to model relationships between selected entities and the surrounding scene.

Full-scene aggregation. We aggregate the contextualized tokens using global average pooling:

$$v_t = \frac{1}{N} \sum_{i=1}^N Z_{t,i}. \quad (6)$$

Unlike an object-only representation, v_t summarizes the complete observation after prompt-guided contextualization. It therefore exposes both the selected regions and the surrounding scene to the downstream policy.

Architecture details. Our default RoboMP-DINOv2 implementation uses DINOv2 ViT-B/14, with backbone dimension $d_b = 768$ and patch size $p = 14$. Input images are resized to 112×112 , producing an 8×8 feature grid. We set the encoder dimension to $d = 256$ and use three pre-normalization transformer encoder layers, each with eight attention heads and a feed-forward dimension of $4d$. The DINOv2 backbone is initialized from pretrained weights and jointly fine-tuned with the projection layer, prompt embeddings, positional embedding, contextualization transformer, output normalization, and downstream policy.

2.3 Policy Conditioning and Optimization

We encode the language instruction using a frozen CLIP text encoder [Radford et al., 2021]. Its normalized output is projected using a learned multilayer perceptron h_{lang} . At each time step, the policy-conditioning feature is

$$c_t = v_t \parallel q_t \parallel h_{\text{lang}}(\text{normalize}(\text{CLIP}(\ell))), \quad (7)$$

where $\parallel$ denotes concatenation.

For an observation horizon T_o , we concatenate the most recent conditioning features,

$$C_t = c_{t-T_o+1} \parallel \cdots \parallel c_t, \quad (8)$$

and provide C_t as global conditioning to Diffusion Policy [Chi et al., 2023].

Let A_t denote a demonstrated action sequence, k a diffusion step, and $\epsilon \sim \mathcal{N}(0, I)$ Gaussian noise. Following the standard forward diffusion process, the noisy action sequence is

$$A_t^{(k)} = \sqrt{\bar{\alpha}_k} A_t + \sqrt{1 - \bar{\alpha}_k} \epsilon, \quad (9)$$

where $\bar{\alpha}_k$ is determined by the diffusion noise schedule. The noise-prediction network ϵ_θ is optimized using

$$\mathcal{L}_{\text{policy}} = \mathbb{E}_{t,k,\epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(A_t^{(k)}, k, C_t \right) \right\|_2^2 \right]. \quad (10)$$

We introduce no auxiliary feature-alignment or contrastive objective. RoboMP-DINOv2 and the downstream Diffusion Policy are optimized jointly using only the policy training objective in Eq. 10.

2.4 Masked-Region Color Randomization

Feature-space prompting identifies selected regions but does not explicitly prevent the encoder from exploiting correlations between their appearance and the demonstrated actions. To improve generalization to unseen object colors, we introduce *masked-region color randomization* (MCR) as an optional training augmentation for RoboMP-DINOv2.

Let $\mathcal{R}_{\text{aug}} \subseteq \{1, \dots, R\}$ denote the mask slots selected for randomization. We construct their union as

$$M_t^{\text{aug}} = \text{clip} \left(\sum_{r \in \mathcal{R}_{\text{aug}}} M_t^{(r)}, 0, 1 \right). \quad (11)$$

For each training image, we sample a spatially constant RGB color $a_t \sim \mathcal{U}([0, 1]^3)$ and replace the pixels inside the union mask:

$$\tilde{I}_t = (1 - M_t^{\text{aug}}) \odot I_t + M_t^{\text{aug}} \odot a_t, \quad (12)$$

where masks and colors are broadcast over the channel and spatial dimensions as needed. Equation 12 is expressed in raw RGB space; in implementation, we apply the corresponding transformation consistently with the input normalization used by DINOv2.

The randomized image $\tilde{I}_t$ is passed through the visual backbone, while the original binary masks are retained for feature-space prompting in Eq. 4. In our experiments, a single sampled color is applied to the union of the selected regions in each image. MCR therefore changes the appearance of the selected regions while preserving their spatial support, the scene geometry, and all unmasked visual context. MCR is applied only during training; evaluation uses the original RGB observations.

We refer to the encoder trained without this augmentation as **RoboMP-DINOv2**. When MCR is enabled during training, we denote the complete encoder-and-augmentation pipeline as **RoboMP-DINOv2-MCR**. The two variants share the same inference architecture; MCR changes only the training-time RGB augmentation.

3 Experiments

Our experiments evaluate the proposed approach under three forms of visual distribution shift. We first evaluate *RoboMP-DINOv2* under held-out spatial configurations and added scene clutter. We then evaluate the *RoboMP-DINOv2-MCR* training pipeline under unseen task-object appearances. Finally, we compare RoboMP-DINOv2 with closely matched architectural variants and analyze how strongly the learned representations respond to the evaluated visual perturbations.

3.1 Experimental Setup

Task suite. We consider seven simulated manipulation settings (Fig. 1): Place Cube, Put Can, Put Can with an obstacle, Pull Tool, Peg Insertion, Long Horizon, and Two-Arm cube stacking. Together, these settings cover pick-and-place, obstacle avoidance, contact-rich insertion, tool use, multi-stage interaction, and bimanual coordination.

The two Put Can variants are particularly useful for evaluating full-scene representations. The obstacle-free and obstacle-present variants share the same high-level goal but may require different motion trajectories. The policy receives no explicit obstacle indicator and must infer from the visual observation whether an obstacle blocks the direct path. A representation that retains only the masked can and target regions may therefore omit information needed to select the appropriate trajectory, whereas RoboMP-DINOv2 preserves the surrounding visual context.

Evaluation regimes. All policies are trained using demonstrations collected in visually clean scenes. We evaluate three out-of-distribution (OOD) regimes:

- **Clean Spatial OOD:** Task objects and goals appear at held-out positions, while the surrounding scene retains the clean visual appearance observed during training.

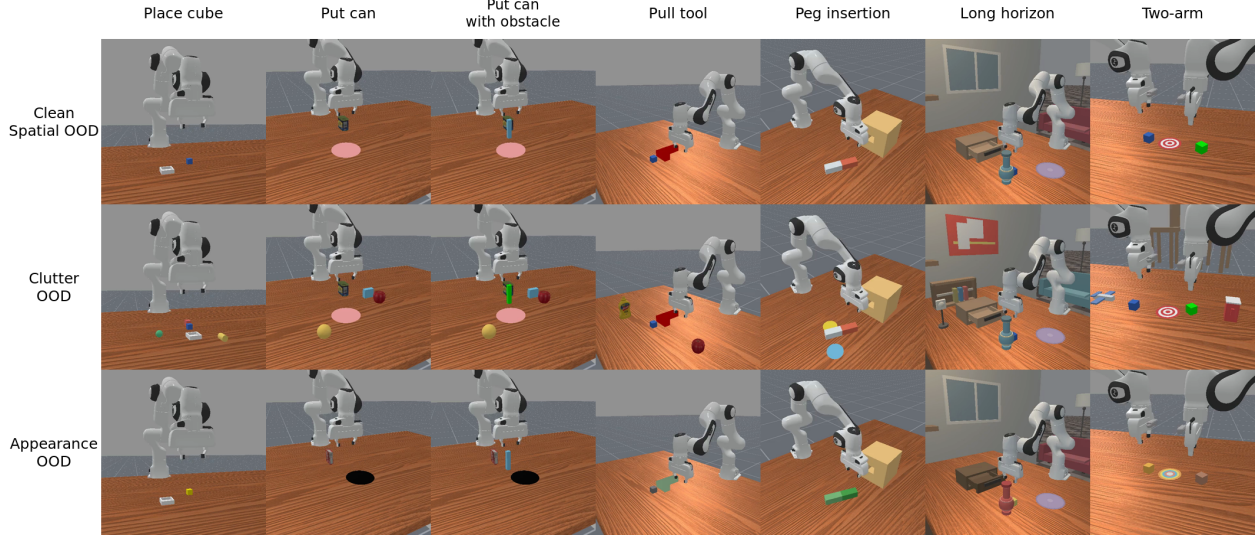


Figure 1: Seven simulated manipulation settings under the three evaluation regimes. Columns denote task settings, and rows show representative observations from Clean Spatial OOD, Clutter OOD, and Appearance OOD.

- **Clutter OOD:** Additional task-irrelevant objects are inserted into the scene. Behaviorally relevant obstacles, when present as part of the task, remain relevant and must not be ignored.
- **Appearance OOD:** Task objects are rendered using colors not observed in the training demonstrations. For selected settings, objects are also initialized at held-out positions, combining appearance and spatial variation.

Clean Spatial OOD is already out of distribution with respect to object configuration and should therefore not be interpreted as an in-distribution reference condition.

Baselines. We compare RoboMP-DINOv2 with four representative visual-representation baselines. **2D object-centric masking** follows the image-space decomposition of DOCIR [Emukpere et al., 2025]. It retains pixels corresponding to the selected objects and robot arm while replacing the remaining image regions with a uniform white background. **3D object-centric masking** follows the object-level point-cloud representation of GROOT [Zhu et al., 2023]. It retains 3D points corresponding to the selected objects and robot arm while filtering the remaining scene points. **DP-DINOv2** combines a standard DINOv2 visual encoder [Hou et al., 2024] with Diffusion Policy [Chi et al., 2023]. **RoboGround-DINOv2** adapts the grounded mask-conditioning mechanism of RoboGround [Huang et al., 2025] to the DINOv2 policy setting by concatenating the masks with the RGB observation. We denote our encoder by **RoboMP-DINOv2** and its training-time MCR variant by **RoboMP-DINOv2-MCR**. For all DINOv2-based methods, the visual backbone is jointly fine-tuned with the downstream policy.

Architecture and comparison protocol. All methods use the same 100 demonstrations per task, observation history, proprioceptive and language inputs, downstream Diffusion Policy, optimization objective, and training budget. The methods differ only in their visual representations and method-specific preprocessing. For each baseline, we preserve its default representation design rather than forcing all visual encoders to use the same architecture or parameter budget.

All mask-based methods receive identical mask inputs: simulator ground-truth masks during training and the same Grounding DINO+SAM predictions [Ren et al., 2024] during evaluation. This protocol controls for segmentation quality and focuses the comparison on how each representation uses the available RGB and mask information.

Metrics. Our primary metric is task success rate, defined as the fraction of evaluation rollouts satisfying the task-specific success criterion. We additionally analyze the learned visual representations using paired clean and perturbed observations that share the same underlying simulator state. Given a clean feature vector $\mathbf{z}$ and its perturbed counterpart $\tilde{\mathbf{z}}$, we compute cosine distance and relative ℓ_2 distance:

$$d_{\cos} = 1 - \frac{\mathbf{z}^\top \tilde{\mathbf{z}}}{\|\mathbf{z}\|_2 \|\tilde{\mathbf{z}}\|_2}, \quad d_{\text{rel}} = \frac{\|\mathbf{z} - \tilde{\mathbf{z}}\|_2}{\|\mathbf{z}\|_2 + \epsilon}. \quad (13)$$

Both metrics normalize for feature magnitude and can be applied to representations with different dimensionalities. Lower values indicate that the evaluated visual perturbation produces a smaller representation change when the underlying simulator state is fixed. These metrics diagnose perturbation sensitivity; by themselves, they do not establish whether a representation preserves all information required for control.

Scope of empirical claims. Our primary objective is to evaluate RoboMP-DINOv2 and RoboMP-DINOv2-MCR as complete visual-representation designs rather than to estimate the causal effect of an individual architectural or augmentation component. The main comparisons therefore report the observed performance of complete encoder and preprocessing pipelines. We additionally evaluate closely matched architecture and augmentation variants to contextualize the proposed designs. We interpret these comparisons as controlled empirical diagnostics rather than causal estimates of the contribution of any single component.

3.2 RoboMP-DINOv2 under Spatial and Clutter Shifts

We first evaluate RoboMP-DINOv2 without masked-region color randomization.

3.2.1 System-Level Evaluation

Evaluation scope. Table 1 compares the complete RoboMP-DINOv2 architecture with alternative visual representations under held-out spatial configurations and added scene clutter. These experiments evaluate RoboMP-DINOv2 as a complete representation design, including dense full-scene feature extraction, multi-region mask prompting, global contextualization, and scene-level aggregation.

Clean Spatial OOD. RoboMP-DINOv2 achieves a seven-setting macro-average success rate of 60.7%, compared with 55.9% for RoboGround-DINOv2 and 50.7% for DP-DINOv2. It achieves the highest observed success on six of the seven settings, with the 3D object-centric baseline performing best on Place Cube. These results demonstrate the effectiveness of the complete RoboMP-DINOv2 architecture under held-out spatial configurations.

The results also reveal a task-dependent pattern among the object-centric baselines. The 3D object-centric baseline performs well on Place Cube, where the information needed for control is concentrated within a small number of selected objects. Its performance decreases substantially on settings involving obstacle avoidance, tool use, insertion, long-horizon interaction, and bimanual coordination. Similar limitations appear for the 2D object-centric and RoboGround-DINOv2 baselines when information outside the selected regions becomes important.

Table 1: Success rates under held-out spatial configurations and scene clutter. All methods are trained on demonstrations collected in visually clean scenes. RoboMP-DINOv2 denotes the proposed mask-prompted full-scene encoder without masked-region color randomization. “Avg.” is the macro-average over the seven settings. The highest observed performance in each column and regime is shown in bold.

Method	Place cube	Put can (clear)	Put can (obstacle)	Pull tool	Peg insertion	Long horizon	Two-arm	Avg.
<i>Clean Spatial OOD</i>								
2D object-centric masking	.350	.190	.000	.000	.020	.007	.000	.081
3D object-centric masking	.850	.840	.070	.000	.085	.036	.120	.286
DP-DINOv2	.410	.820	.750	.150	.170	.812	.440	.507
RoboGround-DINOv2	.690	.820	.770	.000	.215	.789	.630	.559
RoboMP-DINOv2	.540	.860	.830	.170	.230	.848	.770	.607
<i>Clutter OOD</i>								
2D object-centric masking	.310	.120	.000	.000	.000	.009	.000	.063
3D object-centric masking	.780	.260	.030	.000	.000	.018	.060	.164
DP-DINOv2	.340	.590	.520	.130	.130	.751	.410	.410
RoboGround-DINOv2	.640	.600	.030	.010	.030	.595	.670	.368
RoboMP-DINOv2	.560	.830	.800	.180	.210	.787	.810	.597

Clutter OOD. Under added scene clutter, RoboMP-DINOv2 achieves a macro-average success rate of 59.7%, compared with 41.0% for DP-DINOv2 and 36.8% for RoboGround-DINOv2. On obstacle-aware can placement, RoboMP-DINOv2 achieves 80.0% success, compared with 52.0% for DP-DINOv2 and 3.0% for the 3D object-centric baseline. In this setting, information outside the selected object masks—particularly the obstacle and its relationship to the surrounding scene—can directly affect the required trajectory. Hard object-centric filtering can therefore remove information needed for control.

Manually detecting and masking every possible obstacle or distractor would require a rigid, task-specific perception pipeline. RoboMP-DINOv2 instead retains the complete scene and uses the supplied masks only as spatial prompts. Its contextualization layers allow prompted objects to be represented jointly with unprompted obstacles, distractors, and other scene content. RoboMP-DINOv2 also achieves 81.0% success on Two-Arm stacking, compared with 41.0% for DP-DINOv2.

Imperfect predicted masks. The qualitative examples in Fig. 4 show that masks predicted by Grounding DINO and SAM are not always perfectly aligned with the selected objects. This reflects a practical setting in which evaluation-time segmentation cannot be assumed to be exact. In hard object-centric representations, mask errors can directly discard visual evidence by removing regions outside the predicted masks. In RoboMP-DINOv2, mask errors alter the spatial prompting signal but do not delete the underlying full-scene features. Although these examples do not establish explicit robustness to controlled mask corruption, they illustrate how RoboMP-DINOv2 avoids making predicted masks a hard information bottleneck.

3.2.2 Matched Architecture Comparisons

The preceding experiments compare complete encoder designs. To examine RoboMP-DINOv2 relative to closely related architectural alternatives, we compare it with three controlled variants on representative tasks, as shown in Fig. 2:

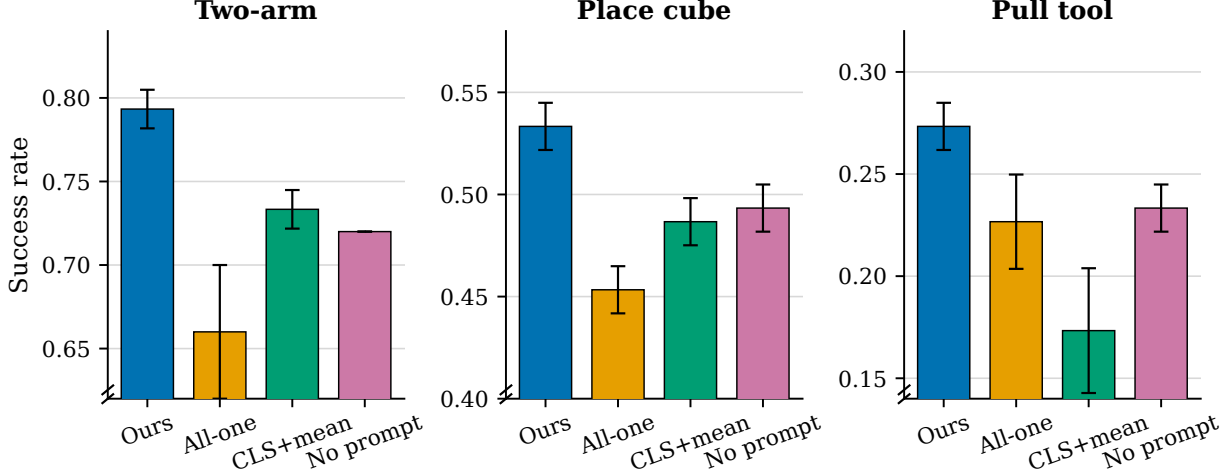


Figure 2: Matched architecture comparison under cluttered-scene evaluation. Bars show mean success over three training seeds evaluated on the same 50 initial conditions; error bars show the standard deviation across seeds. All variants produce a 256-dimensional visual representation and use the same downstream policy. All-one masks retain prompting but remove spatial localization. No mask prompting removes the additive prompt term while retaining the remaining RoboMP-DINOv2 architecture. DINOv2 CLS+mean uses the global DINOv2 class token and mean-pooled patch features without the contextualization transformer.

- **All-one masks** replaces every segmentation mask with an all-one mask, causing the learned prompt embeddings to be added uniformly to all patch tokens. This variant retains the prompt parameters and full-scene contextualization transformer while removing the spatial localization supplied by the masks.
- **No mask prompting** retains the DINOv2 backbone, projection layer, positional embedding, three-layer transformer, normalization, and global pooling used by RoboMP-DINOv2 but removes the additive prompt term in Eq. 4.
- **DINOv2 CLS+mean** concatenates the DINOv2 class token with the mean-pooled patch token and maps the resulting 1536-dimensional feature to 256 dimensions using a linear layer, LayerNorm, and GELU.

Results. RoboMP-DINOv2 achieves the highest observed mean success across all three evaluated settings. The all-one-mask and no-prompt variants provide closely matched reference points in which spatial localization or additive prompting is removed while most of the remaining encoder architecture is retained. The DINOv2 CLS+mean variant provides a conventional global-readout reference. Together, these comparisons support RoboMP-DINOv2 as a complete encoder architecture in the evaluated settings. We interpret the differences as comparative evidence about the complete architectural variants rather than as causal estimates of any individual component.

3.2.3 Feature Consistency under Clutter

We next examine how strongly added clutter changes the learned representation when the underlying simulator state is fixed. Table 2 reports the distances between paired clean and perturbed features. RoboMP-DINOv2 produces lower cosine and relative ℓ_2 distances than DP-DINOv2 on four of the five evaluated settings. The exception is Two-Arm stacking, for which DP-DINOv2 produces slightly lower distances under clutter.

Table 2: Feature distances between perturbed observations and their paired clean counterparts (mean $\pm$ standard deviation over N observation pairs). Cosine distance is reported in units of 10^{-3} and relative ℓ_2 distance in units of 10^{-2} . DP-DINOv2 uses a 768-dimensional feature, whereas RoboMP-DINOv2 uses a 256-dimensional feature. Under Clutter, RoboMP-DINOv2 denotes the encoder trained without color randomization. Under Appearance, DP-DINOv2 is trained without color randomization, whereas RoboMP-DINOv2-MCR denotes the complete encoder-and-augmentation pipeline. Lower values indicate smaller representation changes under the evaluated perturbation; the lower mean is shown in bold.

Task	Shift	N	Cosine $\downarrow$		Relative ℓ_2 $\downarrow$	
			DP-DINOv2	RoboMP-DINOv2 / -MCR	DP-DINOv2	RoboMP-DINOv2 / -MCR
Out-of-bin	Clutter	1269	31.770 $\pm$ 29.030	2.880$\pm$2.990	22.938 $\pm$ 10.619	6.707$\pm$3.555
	Appearance	1269	21.752 $\pm$ 30.249	.0028$\pm$.0064	16.784 $\pm$ 12.520	.181$\pm$.150
Pull tool	Clutter	2409	.002 $\pm$.005	.001$\pm$.002	.153 $\pm$.109	.126$\pm$.077
	Appearance	2409	.240 $\pm$.485	.007$\pm$.037	1.856 $\pm$ 1.509	.230$\pm$.281
Peg insertion	Clutter	1514	.625 $\pm$ 1.347	.249$\pm$.546	2.642 $\pm$ 2.373	1.636$\pm$1.516
	Appearance	1514	.553 $\pm$ 1.621	.013$\pm$.042	2.073 $\pm$ 2.634	.375$\pm$.358
Long horizon	Clutter	4740	.105 $\pm$.115	.040$\pm$.091	1.296 $\pm$.700	.665$\pm$.601
	Appearance	4740	.216 $\pm$.313	.002$\pm$.005	1.658 $\pm$ 1.322	.170$\pm$.138
Two-arm	Clutter	3028	1.769$\pm$1.315	2.025 $\pm$ 1.576	5.595$\pm$2.141	5.874 $\pm$ 2.449
	Appearance	3028	13.588 $\pm$ 27.961	.019$\pm$.072	11.813 $\pm$ 15.161	.371$\pm$.494

These results show that, under the evaluated training configurations, RoboMP-DINOv2 is generally less sensitive than DP-DINOv2 to the addition of scene clutter. This analysis characterizes the representations produced by the complete encoders; it does not isolate the contribution of an individual RoboMP-DINOv2 component.

3.3 Appearance Generalization with Masked-Region Color Randomization

We next evaluate robustness to unseen task-object colors. We first use the Appearance rows of Table 2 to measure how strongly recoloring the task objects changes the learned representation when the underlying simulator state is fixed. DP-DINOv2 is trained without color randomization in this representation-level comparison. Its features change substantially under task-object recoloring, even though the scene geometry and required action remain unchanged. In contrast, the complete RoboMP-DINOv2-MCR pipeline reduces relative ℓ_2 distance by $92.9\times$ on Out-of-bin, $8.1\times$ on Pull Tool, $5.5\times$ on Peg Insertion, $9.8\times$ on Long Horizon, and $31.8\times$ on Two-Arm stacking. The same overall trend appears in cosine distance.

This comparison shows that, under the evaluated training configurations, task-object recoloring produces substantially smaller representation changes for RoboMP-DINOv2-MCR than for unaugmented DP-DINOv2. Because the two pipelines differ in both encoder architecture and training augmentation, the comparison characterizes their resulting perturbation sensitivity without attributing the difference to either component individually.

Table 3 provides a system-level behavioral comparison in which every method is trained with color randomization. Because the standard DP-DINOv2 baseline does not use task masks, its randomization is applied to the complete image. For mask-based methods, randomization is restricted to the selected mask regions. RoboMP-DINOv2-MCR achieves the highest observed success in all seven settings, tying RoboGround-DINOv2 on Place Cube and outperforming it on the remaining six. Its seven-setting macro-average is 72.5%, compared with 35.1% for RoboGround-DINOv2 and 19.1% for DP-DINOv2. Under this comparison protocol, RoboMP-DINOv2-MCR achieves the strongest overall performance among the

Table 3: Success rates under Appearance OOD. All methods are trained with color randomization. DP-DINOv2 uses full-image randomization, whereas the mask-based methods apply randomization only within selected mask regions. “Avg.” is the macro-average over the seven settings. The highest observed performance in each column is shown in bold.

Method	Place cube	Put can (clear)	Put can (obstacle)	Pull tool	Peg insertion	Long horizon	Two-arm	Avg.
2D object-centric masking (masked-area rand.)	.790	.040	.010	.000	.000	.003	.000	.120
3D object-centric masking (masked-area rand.)	.940	.030	.000	.000	.010	.015	.000	.142
DP-DINOv2 (full-image rand.)	.330	.050	.100	.100	.250	.498	.010	.191
RoboGround-DINOv2 (masked-area rand.)	.990	.650	.660	.000	.000	.160	.000	.351
RoboMP-DINOv2-MCR	.990	.710	.670	.820	.700	.906	.280	.725

Table 4: Appearance-randomization comparison under combined appearance-and-clutter shifts. Values report mean success rates across the three representative tasks.

Method	Two arm	Place cube	Pull tool
DP + full-image rand.	0.00	0.02	0.00
DP + masked rand.	0.14	0.08	0.34
Only mask + mask rand.	0.00	0.04	0.00
RoboMP-DINOv2-MCR	0.26	0.24	0.38

evaluated complete pipelines.

Robustness under simultaneous appearance and clutter shifts. We next consider a more challenging condition in which appearance and clutter shifts occur simultaneously. Table 4 evaluates Two-Arm stacking with both unseen object colors and additional scene clutter. DP-DINOv2 with full-image randomization obtains a mean score of 0.01. Restricting the augmentation to selected mask regions, without providing the masks to the encoder, improves the score to 0.11. This result indicates that the spatial support of the augmentation is consequential even in the absence of mask-conditioned encoding. Combining masked-region randomization with RoboGround-DINOv2 and 2D object-centric masking produces scores of 0.07 and 0.00, respectively. RoboMP-DINOv2-MCR achieves the highest score, 0.20, among the evaluated combinations.

Preserving behaviorally relevant color cues. We finally consider a context-dependent setting in which the correct action depends on color information outside the randomized regions. Figure 3 shows a kitchen task in which the stove-light color determines the required behavior: a green light instructs the robot to remove the pot from the stove, whereas a red light instructs it to place the plated butter into the pot. Full-image color randomization can alter this decision-relevant cue during training and encourage the policy to ignore it. MCR instead randomizes only the selected object regions, while RoboMP-DINOv2 retains the unmodified scene context.

RoboMP-DINOv2-MCR achieves success rates of 88% and 71% in the green- and red-light conditions, respectively, compared with 25% and 19% for DP-DINOv2 trained with full-image color randomization. This experiment demonstrates the behavior of the complete RoboMP-DINOv2-MCR pipeline in a setting that requires robustness to task-object recoloring while preserving sensitivity to a behaviorally relevant contextual cue. It also illustrates how RoboMP-DINOv2 provides a suitable full-scene architecture for localized augmentation: MCR alters the selected object regions without preventing the encoder from using the unmodified stove-light signal.

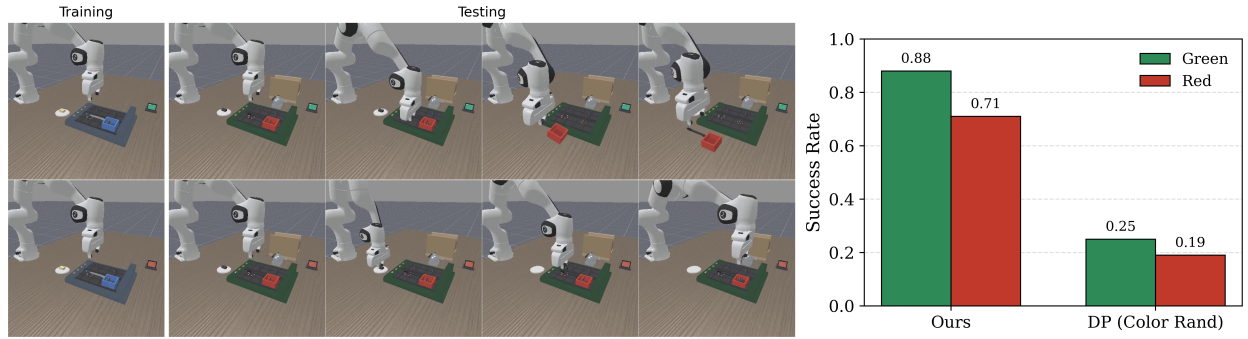


Figure 3: Preserving behaviorally relevant color cues outside the randomized regions. The stove-light color determines the correct action. RoboMP-DINOv2-MCR retains access to this contextual signal and substantially outperforms DP-DINOv2 trained with full-image color randomization under both light conditions.

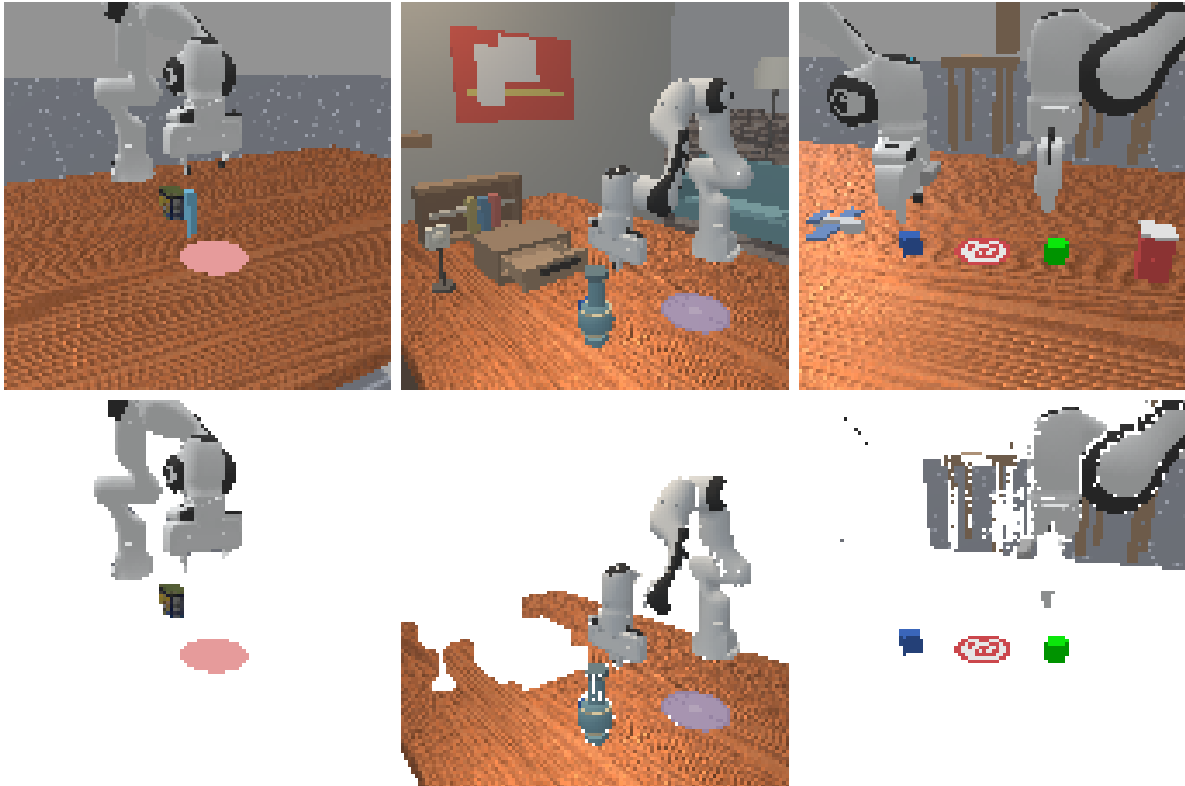


Figure 4: Qualitative examples of imperfect evaluation-time mask predictions from Grounding DINO+SAM. Top row: RGB observations. Bottom row: predicted masks used by all mask-based methods. From left to right, the columns show Put Can (obstacle), Long Horizon, and Two-Arm stacking.

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